THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 7: **Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The distance of the point $(-6, 8)$ from the origin is: 1

(a) 6 **(b)** 8 **(c)** 10 **(d)** 14

2. The distance between $(-5, 7)$ and $(-1, 3)$ is: 1

(a) $4\sqrt{2}$ **(b)** $2\sqrt{2}$ **(c)** 8 **(d)** 4

3. The mid-point of the segment joining $(2, -3)$ and $(6, 5)$ is: 1

(a) $(4, 1)$ **(b)** $(2, 4)$ **(c)** $(8, 2)$ **(d)** $(4, -1)$

4. A point on the x-axis is of the form: 1

(a) $(0, a)$ **(b)** $(a, 0)$ **(c)** (a, a) **(d)** $(a, -a)$

5. A student states: "The point $(0, -5)$ lies on the x-axis." The correct response is: 1

(a) Correct **(b)** Incorrect — it lies on the y-axis

(c) It lies in Quadrant IV **(d)** It is the origin

6. The ratio in which the x-axis divides the segment joining $(2, -3)$ and $(5, 6)$ is: 1

(a) 1 : 2 **(b)** 2 : 1 **(c)** 3 : 5 **(d)** 5 : 3

7. For A(1, 2) and B(4, 6), Student A says $AB = 5$ and Student B says $AB = 7$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. If $\left(\frac{a}{3}, 4\right)$ is the mid-point of the segment joining $(-6, 5)$ and $(-2, 3)$, then a is: 1

(a) −12 **(b)** −4 **(c)** 12 **(d)** −6

9. The point dividing the join of $(1, 3)$ and $(4, 6)$ internally in the ratio 2 : 1 is: 1

(a) $(3, 5)$ **(b)** $(2, 4)$ **(c)** $(3, 4)$ **(d)** $(2, 5)$

10. If the distance between $(4, p)$ and $(1, 0)$ is 5, then p is: 1

(a) ± 4 **(b)** 4 only **(c)** −4 only **(d)** 0

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the distance between the points (a, b) and $(-a, -b)$. 2

12. Find the point on the y-axis which is equidistant from A(6, 5) and B(−4, 3). 2

13. Find the coordinates of the point which divides the join of $(4, -3)$ and $(8, 5)$ internally in the ratio 3 : 1. 2

14. A student claims that A(3, 0), B(6, 4) and C(−1, 3) are the vertices of a right isosceles triangle. Verify. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the ratio in which the line segment joining $(-3, 10)$ and $(6, -8)$ is divided by $(-1, 6)$. 3

16. Find a relation between x and y such that the point (x, y) is equidistant from $(3, 6)$ and $(-3, 4)$. 3

17. Find the coordinates of the points which divide the segment joining A(−2, 2) and B(2, 8) into four equal parts. 3

OR

If A(1, 2), B(4, y), C(x, 6) and D(3, 5) are the vertices of a parallelogram taken in order, find x and y. 3

18. Show that the points $(1, 7)$, $(4, 2)$, $(-1, -1)$ and $(-4, 4)$ are the vertices of a square. 3

OR

Find the ratio in which the y-axis divides the segment joining $(5, -6)$ and $(-1, -4)$. Also find the point of intersection. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Find the coordinates of the points of trisection of the segment joining $(4, -1)$ and $(-2, -3)$. Verify, using the distance formula, that the first trisection point is equidistant from $(4, -1)$ and the second trisection point. 5

OR

Case study: In a classroom, four friends are seated at A(3, 4), B(6, 7), C(9, 4) and D(6, 1). Champa says “ABCD is a square”, Chameli disagrees. 5

(i) Find AB, BC, CD, DA.

(ii) Find the diagonals AC and BD.

(iii) Who is correct? Justify.

20. **Case study:** On a town map drawn on a grid, a school is at S(2, 3) and a hospital at H(10, 9). 5

(i) Find the distance SH.

(ii) A park P is to be built on SH dividing it in the ratio 1 : 3. Find P.

(iii) A market M is at the mid-point of SH. Find M.

(iv) Verify that M is equidistant from S and H.

OR

AB is a diameter of a circle with centre $(2, -3)$ and B is $(1, 4)$. 5

(i) Find the coordinates of A.

(ii) Find the radius of the circle.

(iii) Check whether the point $(9, -3)$ lies on the circle.

END OF QUESTION PAPER Total: 40 Marks